



Lab Task 10

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Subject	SCD

Refactored Code

```
1 package project2;
2
3 import java.util.Scanner;
4
5 abstract class Operation {
6     protected double num1;
7     protected double num2;
8
9     public Operation(double num1, double num2) {
10         this.num1 = num1;
11         this.num2 = num2;
12     }
13
14     public abstract double calculate();
15 }
16
17
18 class Addition extends Operation {
19     public Addition(double num1, double num2) {
20         super(num1, num2);
21     }
22
23     public double calculate() {
24         return num1 + num2;
25     }
26 }
27
28 class Subtraction extends Operation {
29     public Subtraction(double num1, double num2) {
30         super(num1, num2);
31     }
32
33
34     public double calculate() {
35         return num1 - num2;
36     }
37 }
38
39 class Multiplication extends Operation {
40     public Multiplication(double num1, double num2) {
41         super(num1, num2);
42     }
43
44     public double calculate() {
45         return num1 * num2;
46     }
47 }
48
49 class Division extends Operation {
50     public Division(double num1, double num2) {
51         super(num1, num2);
52     }
53
54
55     public double calculate() {
56         if (num2 == 0) {
57             throw new ArithmeticException("Cannot divide by zero");
58         }
59         return num1 / num2;
60     }
61 }
62
63 }
```

```

64
65 public class Calculator {
66     public static void main(String[] args) {
67         Scanner scanner = new Scanner(System.in);
68
69         System.out.println("Simple Calculator");
70         double num1 = getNumberInput(scanner, "Enter the first number: ");
71         double num2 = getNumberInput(scanner, "Enter the second number: ");
72         char operator = getOperatorInput(scanner);
73
74         try {
75             Operation operation = createOperation(num1, num2, operator);
76             double result = operation.calculate();
77             System.out.println("Result: " + result);
78         } catch (IllegalArgumentException | ArithmeticException e) {
79             System.out.println("Error: " + e.getMessage());
80         } finally {
81             scanner.close();
82         }
83     }
84
85     private static double getNumberInput(Scanner scanner, String prompt) {
86         System.out.print(prompt);
87         return scanner.nextDouble();
88     }
89
90     private static char getOperatorInput(Scanner scanner) {
91         System.out.print("Enter the operation (+, -, *, /): ");
92         return scanner.next().charAt(0);
93     }
94
95     private static Operation createOperation(double num1, double num2, char operator) {
96         switch (operator) {
97             case '+': return new Addition(num1, num2);
98             case '-': return new Subtraction(num1, num2);
99             case '*': return new Multiplication(num1, num2);
100             case '/': return new Division(num1, num2);
101             default: throw new IllegalArgumentException("Invalid operator. Please enter +, -, *, or /");
102         }
103     }
104 }
105
106
107
108
109
110
111
112
113
114

```

Output

```
Simple Calculator  
Enter the first number: 10  
Enter the second number: 20  
Enter the operation (+, -, *, /): +  
Result: 30.0
```

